2024035

INSTITUTE OF ENGG. AND TECHNOLOGY,DAVV

B.E 2nd YEAR ,ELECTRONICS AND TELECOMMUNICATIONS(DEC 2023)

ANALOG ELECTRONICS (3ETRC4)

TIME 3HRS **MAX MARKS 60**

NOTE: ATTEMPT ANY TWO PARTS FROM EACH QUESTION.EACH QUESTION IS OF 12 MARKS AND EACH PART IN IT IS OF 06 MARKS. BE PRECISE AND UPTO THE MARK. MAKE SUITABLE ASSUMPTIONS IF NEEDED AND MENTION IT EXPLICITLY.PARTS OF THE SAME PROBLEM MUST BE SOLVED IN CONTINUATION.

1 (A) Find the exact expression for h_{fe} in terms of common base H parameters at low frequencies. From this exact relation obtain approximate expression for h_{fe} .

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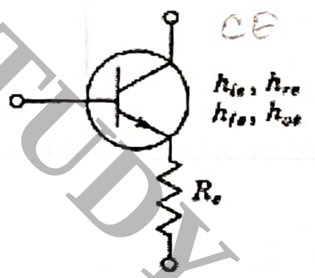
(B) Find the exact expression for h_{rb} in terms of common emitter H parameters at low frequencies. From this exact relation obtain approximate expression for h_{rb} .

(C) For the composite transistor shown below . Derive expression for h'_{ie} for this composite transistor using original H parameters.



2 For the amplifier in the circuit shown below . Take R_s = 0.1Kohms and R_L = 5 kohms . Use standard values of H parameters .

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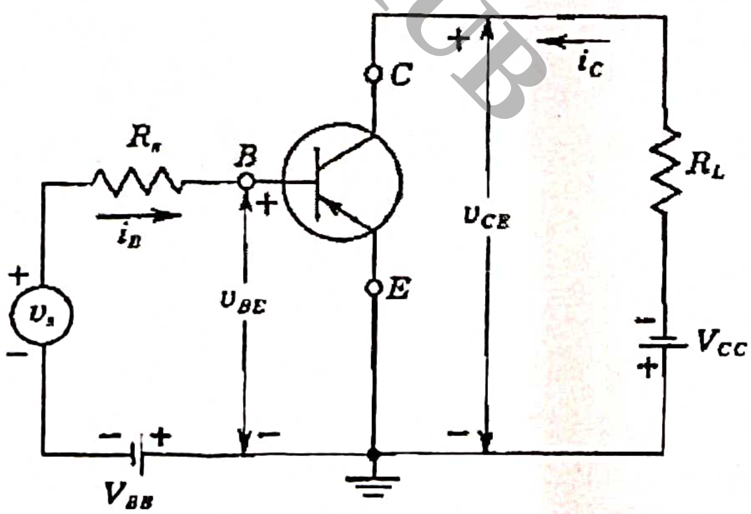
Find out

(A) Current gain A_I

(B) Voltage gain A_V

(C) Input impedance R_i,



3 (A) Derive expression for Input impedance of Darlington's circuit .
(B) Explain the problem of biasing in High input impedance circuits using an example.
(C) Explain method of Bootstrapping to solve biasing problem.

4 (A) Draw Giacolleto model of Bipolar junction transistors. Write down values of its parameters at $I_{CQ} = 1.3$ mA .
(B) Derive expression for g_m at high frequencies for a bipolar junction transistor.
(C) Derive expression for hybrid Pi capacitance C_e .

5 (A) Derive expression for f_β and explain the meaning of f_T . Establish a relation between them.
(B) Draw schematic block diagram of negative feedback based amplifier system. Explain its various components.
(C) Explain phase shift oscillator in full detail.